



December 2025 Updated Analysis of Russian Shahed-type UAVs Deployment Against Ukraine

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Key findings

- In December, the trend of saturation in the number of Shahed-type UAV launches, which had been ongoing since the beginning of autumn, continued. Specifically, 5,131 Shahed-type drones were launched in December, of which 3,065 were designated as strike drones, and the remaining drones were categorized as decoy drones. Furthermore, this is the lowest figure recorded since the onset of autumn. The mean frequency of Shahed drone launches in December was approximately 166 per day. This figure approximates the mean number of Shahed drone launches during the summer and fall of 2025, which was 175 UAVs per day, with a peak in July, when the mean was 203 drones per day (see Fig. 1 and Table 1).
- To summarize, it can be stated that since the onset of summer 2025, the quantity of Shahed drones has remained approximately constant at 5,340 per month, with a maximum of 6,297 in July. This suggests the potential for production saturation or constrained supplies of essential components.
- Presently, there has been an observed increase in the relative number of Shahed-type drones that are equipped with online video cameras and radio modems, as compared to the total number of such UAVs. The implementation of such an upgrade necessitates the utilization of specialized equipment and the presence of operators who have undergone the requisite training to effectively control Shahed-type drones. This phenomenon may also be one of the factors contributing to the observed limitation in the growth of the total number of launches.
- Since September 8, official sources have initiated the publication of the approximate number of Shahed/Geran strike drones in the total number of launches. As demonstrated in Table 1, the mean number of Shahed/Geran strike drones in the final quarter of 2025 approximates 60% of the aggregate number (58% in October, 62% in November, and 60% in December) of drones launched.
 - This finding is consistent with our earlier estimates and facilitates the calculation of the number of Shahed/Geran strike drones produced in Alabuga.
 - Concurrently, it is noteworthy that in mass attacks, the proportion of strike UAVs in the total number of launches typically diminishes, while in targeted attacks

with a limited number of UAVs, the percentage of strike UAVs can reach 80% (see Figure 2).

- A decrease in the total number of Shahed drone launches was observed in December (5,131) compared to November (5,447). However, the absolute number of successful hits in December (886) is higher than in November (826) and ranks second after October (989) for all periods (see Fig. 3). The efficacy of strikes in December was 17.27%, which is the third highest monthly figure for all of 2025 (see Table 1).

Effectiveness of Russian Shahed Strikes

Firstly, it should be emphasized that the main purpose of decoy drones, despite some of them being equipped with small cumulative munitions, is to distract attention and exhaust air defense systems, as well as to conduct video reconnaissance. To assess the effectiveness of strikes, it is imperative to consider not all launched UAVs of the Shahed type, but specifically the strike-type models, such as the Shahed-136/131, Geran2-3, and Harpy 1A. The following assessments will be provided: overall effectiveness and the effectiveness of hits by the strike UAVs (in parentheses).

As demonstrated in Table 1, the average percentage of hits relative to the total number of Shahed-type UAV launches (both strike and decoy drones) exhibited a marked increase from 2-3% in January-February to 9% in March. Since April 2025, this figure has further elevated, ranging from 11.5% to 18.7%, and in terms of the number of strike UAV launches, from 19.5% to 32.3%, respectively. In December, these figures were 17.27% total and 28.21% strike-type UAVs, respectively.

It is noteworthy that during certain attacks, particularly those of a large scale and those that penetrated deep into Ukrainian territory, the relative effectiveness exhibited a decline to 5% total (8% strike UAVs). Conversely, during attacks involving 100–200 drones per day, the effectiveness ranged from 40% to 50% (70%–80%). A notable illustration of this phenomenon occurred on August 9, 2025, when the effectiveness surged remarkably to an exceptional 66%. In December, the maximum hit rate relative to the total number of UAVs launched was 45% on December 11. However, when only strike UAVs are considered, the maximum effectiveness was achieved on December 17 and amounted to 72.5%, as shown in Figures 4.a and 4b. These features must be considered when assessing the effectiveness of strikes by Shahed-type UAVs. At the same time, as demonstrated in Figure 4, Shahed-type UAVs generally exhibit the lowest hit efficiency during mass attacks. In rare instances, this efficiency can also be low during attacks involving a limited number of UAVs. The following is a list of the lowest effectiveness of hits in December:

- 9.2% on December 5-6, with a total number of launches of 653 (20% of 300 strike UAVs);
- 17.8% on December 12-13, with a total number of launches of 460 (10.3% of 270 strike UAVs);
- 7.3% on December 13-14, with a total number of launches of 138 (11.76% of 85 strike UAVs);

- 6.1% on December 22-23, with a total number of launches of 635 (9.75% of 400 strike UAVs); and
- 4.8% on December 26-27, with a total number of launches of 519 (8.33% of 300 strike UAVs).

As noted in the analysis by @ShahedTracker¹, most strikes are concentrated in Kyiv and the adjacent Russian-occupied territory. At the same time, four combined massive attacks in December 2025 focused primarily on energy infrastructure in various regions of Ukraine (see Fig. 5).

Conclusions on the Increase in Effectiveness

Offensives comprising a substantial number of Shahed-type UAVs (>400) are predominantly components of integrated attacks encompassing diverse categories of missiles (refer to Figure 5). These occurrences manifest 4-5 times per month, commencing in the summer of 2025, apart from August, where it was 3 times (refer to Figure 1). In these attacks, they serve a dual purpose: to hit targets, but more importantly, to exhaust Ukraine's air defenses, thereby increasing the effectiveness of missiles, which are significantly more expensive.

A thorough examination of the statistics and geography of strikes involving Shahed-type UAVs reveals the primary factors contributing to the enhanced effectiveness of these strikes within Ukrainian territory:

- First, the proximity of the strike areas to the front line or the Ukrainian border (e.g., Sumy, Chernihiv, Kharkiv, Dnipro, Zaporizhzhia, Odesa, etc.) reduces flight time and allows for more efficient route planning.
- Secondly, recent advancements in navigation systems, particularly the incorporation of 16-element CRPA antennas, have enhanced the capacity to counteract Ukrainian electronic warfare systems.
- Thirdly, the integration of online cameras and modems facilitates real-time adjustment of flight trajectories and evasive maneuvers.

There has been an increase in reports of the use of Shahed/Geran jet-powered strike drones, which are much more difficult to neutralize than conventional Shahed-136/131 drones. However, there have been reports of their successful interception by Ukrainian Sting interceptor drones.²

The most salient factor is the constant use of many Shahed-type drones, which rapidly deplete air defense resources. To address this challenge, it is imperative to fortify countermeasures and foster the development of Ukrainian capabilities to counter these threats, including the deployment of interceptor drones. Additionally, it is crucial to augment the supply of Western air defense systems to Ukraine to effectively combat the threat posed by drones.

¹ <https://x.com/ShahedTracker>

² https://x.com/fab_hinz/status/1995043510597333457?s=20

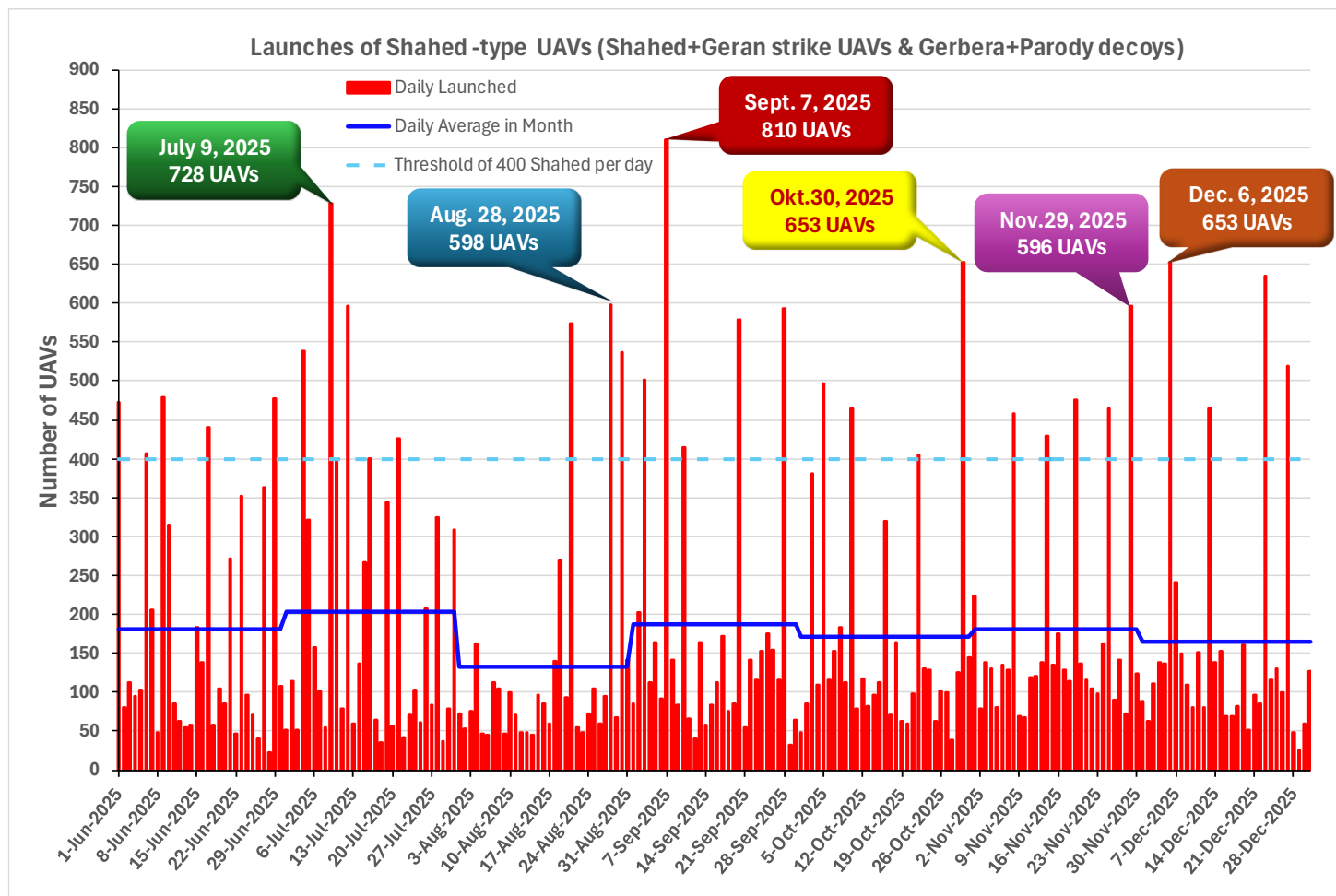


Figure 1. Statistics of the launches of the Shahed-type UAVs during June through December 2025.

	Hits		Interceptions		Launches Total		Launches strike UAVs Shahed & Geran			Hits/ Launches Total	Hits/ strike UAVs launches
	Total	Average daily	Total	Average daily	Total	Average daily	Total official published	Total estimated	% of total	Average monthly	Average monthly
2025	7263		47275		54538		12728	19481	59%		
January	90	2.90	2539	81.90	2629	84.81		1551		3.42%	5.80%
February	102	3.64	3802	135.79	3904	139.43		2303		2.61%	4.43%
March	377	12.16	3821	123.26	4198	135.42		2477		8.98%	15.22%
April	379	12.63	2043	68.10	2422	80.73		1429		15.65%	26.52%
May	714	23.03	3289	106.10	4003	129.13		2362		17.84%	30.23%
June	760	25.33	4678	155.93	5438	181.27		3208		13.98%	23.69%
July	723	23.32	5574	179.81	6297	203.13		3715		11.48%	19.46%
August	680	21.94	3448	111.23	4128	133.16		2436		16.47%	27.92%
September	737	24.57	4906	163.53	5643	188.10	3245		58%	13.06%	22.71%
October	989	31.90	4309	139.00	5298	170.90	3063		58%	18.67%	32.29%
November	826	27.53	4621	154.03	5447	181.57	3355		62%	15.16%	24.62%
December	886	28.58	4245	136.94	5131	165.52	3065		60%	17.27%	28.91%

Table 1. Detailed statistics of the launches, interceptions and hits of the Shahed-type UAVs in 2025.

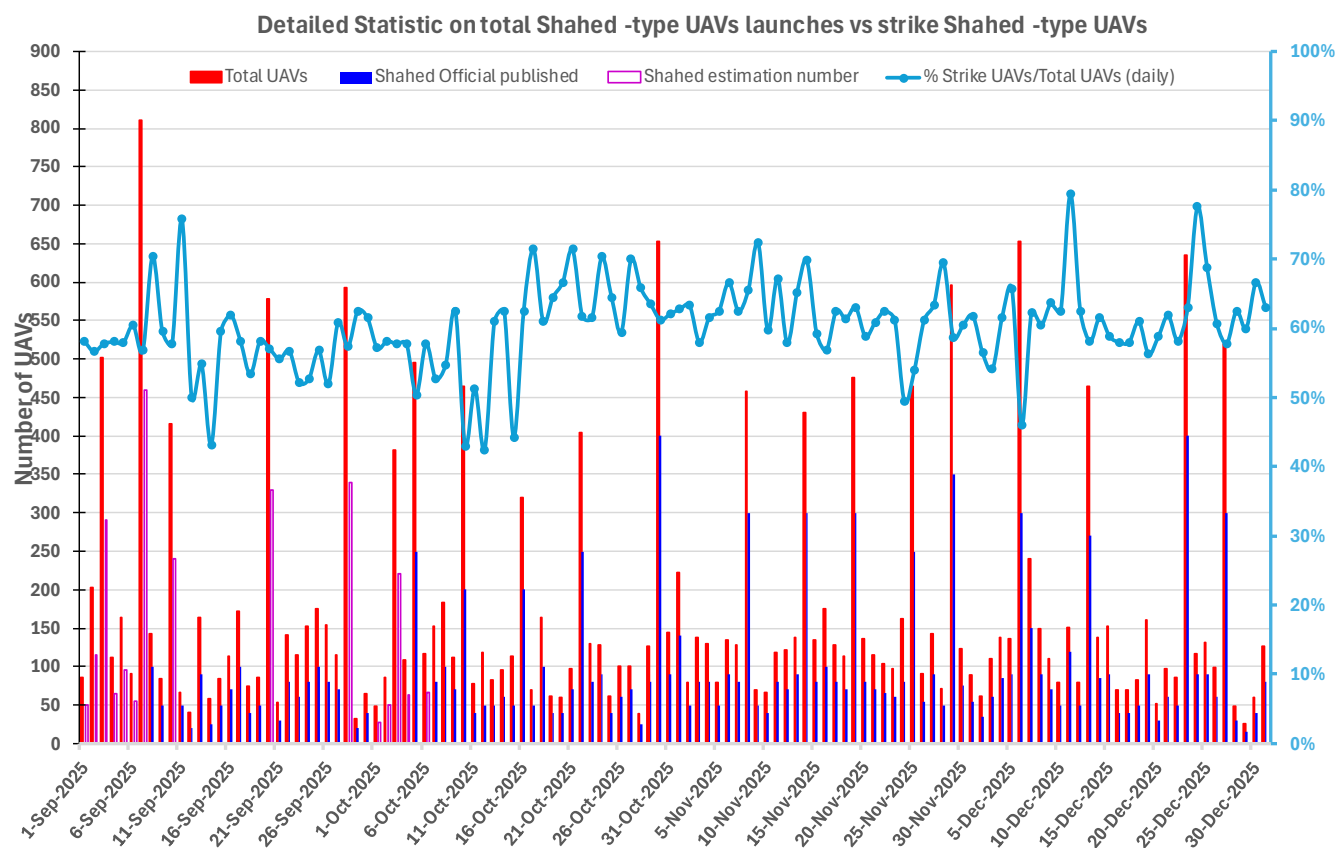


Figure 2. Statistics of the launches of the Shahed+Geran vs total number of strike+decoy UAVs in September through December 2025.

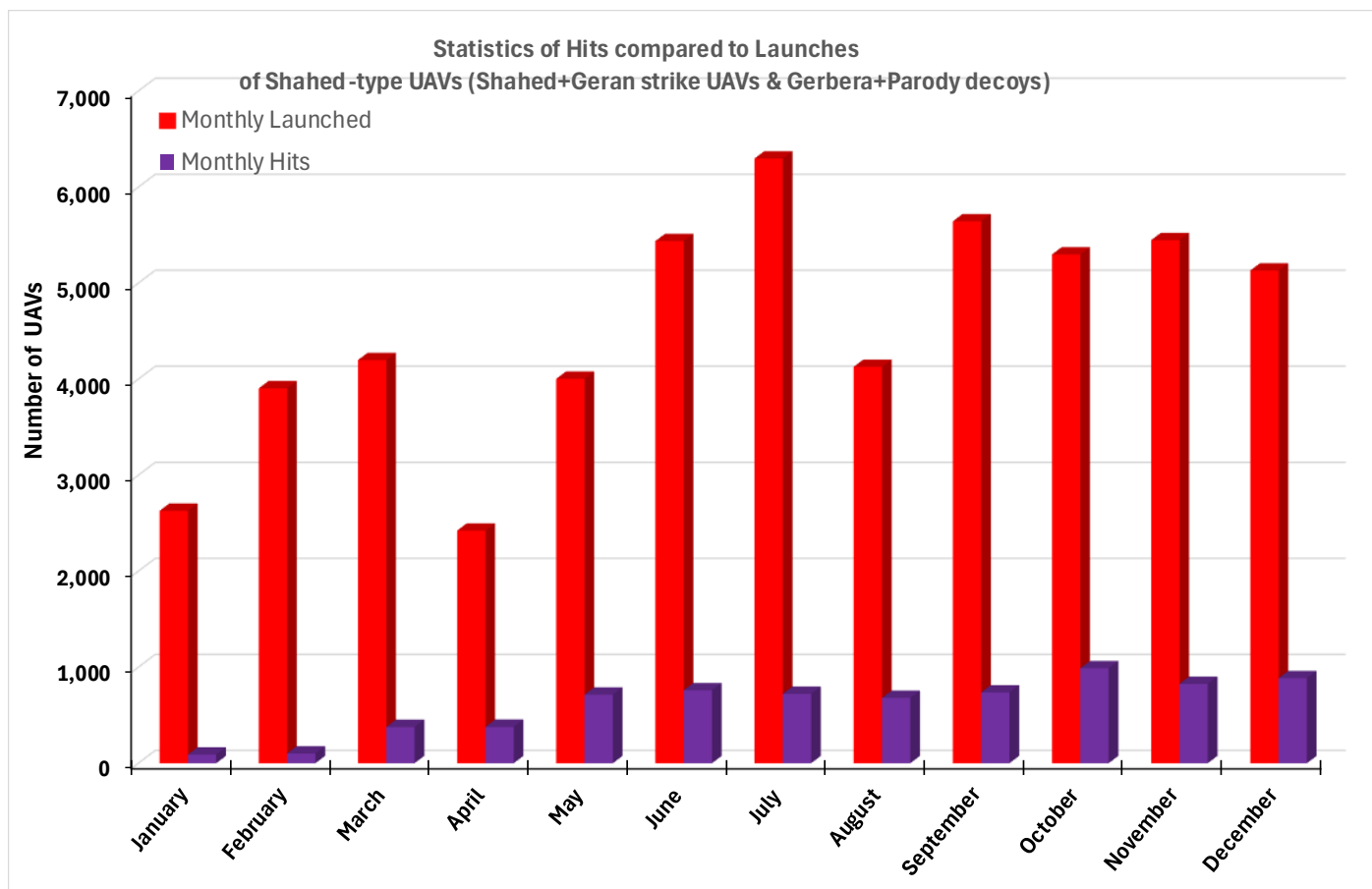


Figure 3. Monthly Hits vs Launches of the Shahed-type UAVs during 2025.

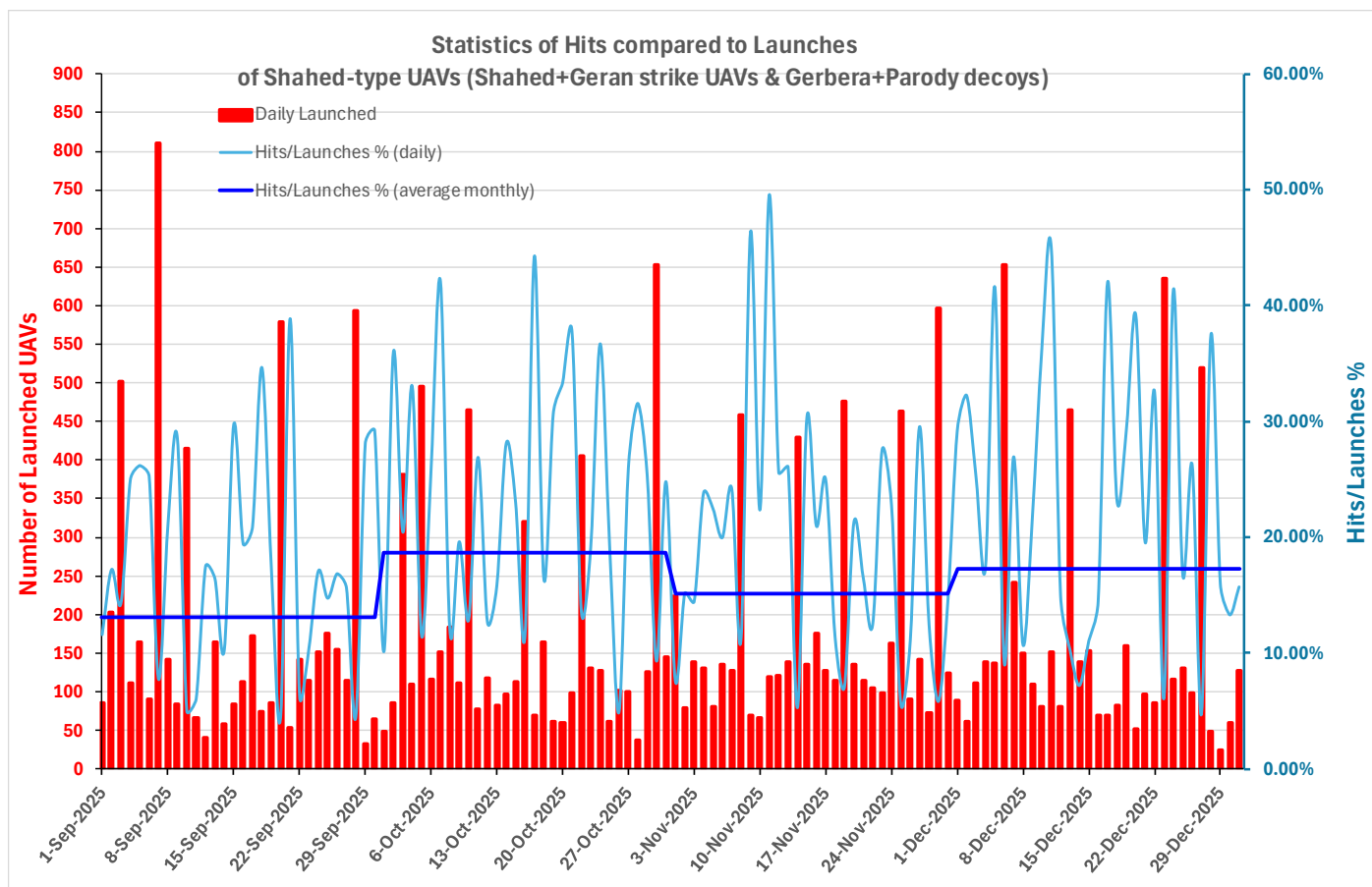


Figure 4a. Statistics of the Hits vs Launches of the total Shahed-type UAVs in September through December 2025.

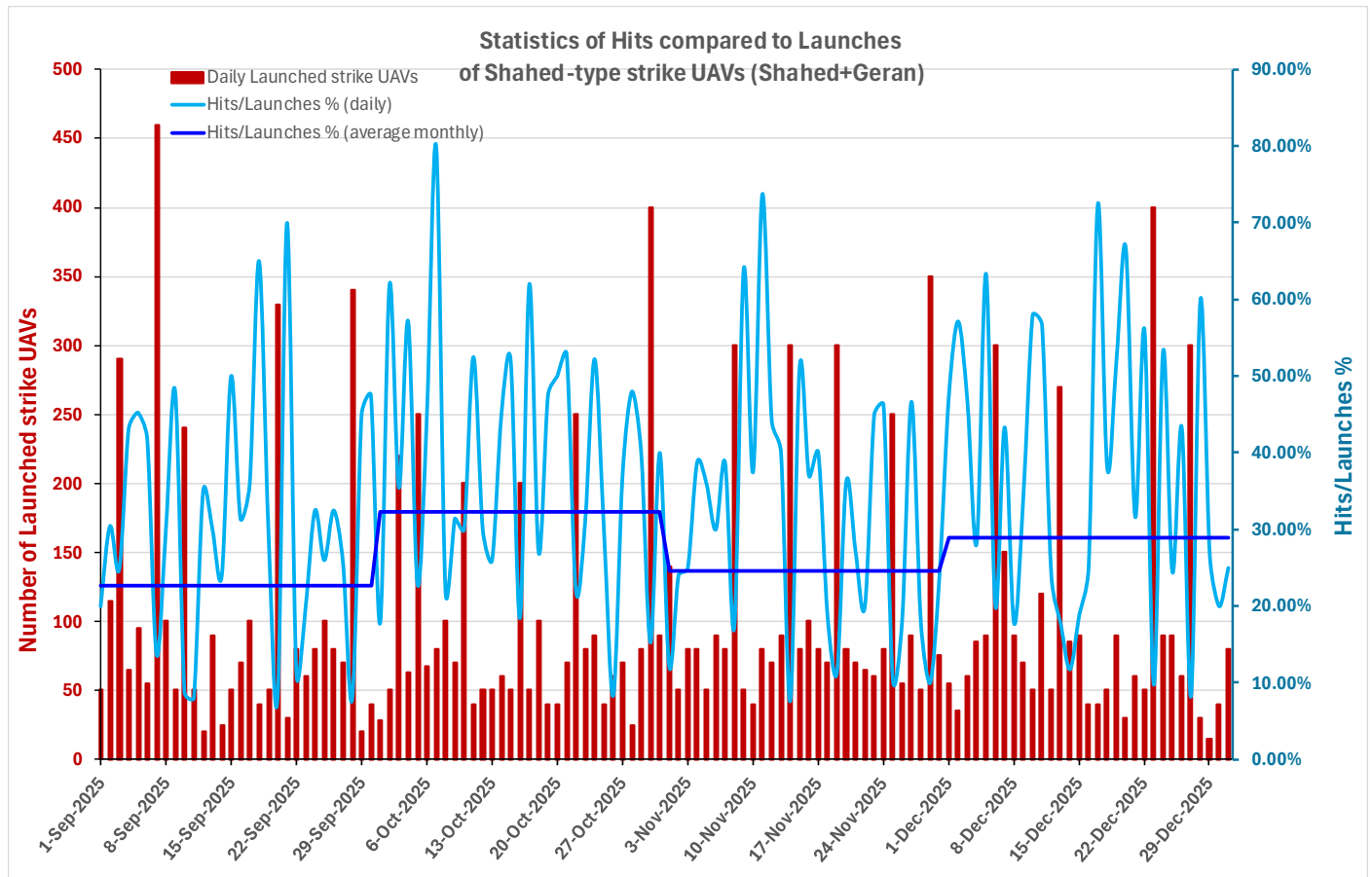


Figure 4b. Statistics of the Hits vs Launches of the Shahed-type strike UAVs in September through December 2025.

